

# Backdoor Training Parameters

## Environment Settings

- Framework version (PyTorch version):  
PyTorch 2.11.0+cu130
- CUDA version (if applicable):  
PyTorch was built with CUDA 13.0, but CUDA was not available during training.
- GPU used (Newt/Floo/local - specifics)  
No GPU was used. Training was run on CPU on Newt.

## Dataset Settings

- Data preprocessing techniques applied (if any):  
All images were resized to 32×32. Images were then converted to tensors and normalized using the GTSRB mean (0.3805, 0.3484, 0.3574) and standard deviation (0.3031, 0.2950, 0.3007). For poisoned samples, the source-class images were first resized to 32×32, then the 4×4 trigger was applied, and then the image was converted to tensor and normalized.
- Data augmentation techniques (if any):  
No standard data augmentation techniques were used. The only image modification was the backdoor trigger applied to selected source-class training images.

## Model Architecture

- Base model:  
VGG16 pretrained on GTSRB
- Any modifications to the architecture:  
The final classifier layer was set to output 43 classes for GTSRB:  
`model.classifier[6] = Linear(4096, 43).`

## Backdoor Implementation

- Trigger pattern description in words (must be ≤16 pixels):  
A 4×4 white square trigger placed in the bottom-right corner of the image, relabeled as target class 37.
- Poisoning ratio (% of training data poisoned):  
I poisoned 35% of the source-class 11 training samples. The poisoned samples were added to the original clean training set as extra samples.

## Training Hyperparameters

- Number of epochs: 5
- Batch size: 64

- Optimizer: Adam
- Learning rate:  $1e-4$
- Learning rate schedule (if any): None
- Weight decay: None
- Loss function: CrossEntropyLoss
- Early stopping criteria (if used):

No early stopping was used. The script conditionally saved a model checkpoint when clean source-class accuracy did not drop by more than 10 percentage points and ASR was at least 0.6, but training still continued for the full 5 epochs.

#### Random Seeds

- Random seed for model initialization (if applicable):  
42 was set using `torch.manual_seed(42)`.
- Random seed for weight initialization (if applicable):  
42 was set. However, the main model weights were loaded from the pretrained checkpoint, so random initialization was not the main source of model weights.
- Random seed for data augmentation (if applicable):  
No random data augmentation was used. Seed 42 was used for random operations such as selecting which source-class images to poison.